

$$\begin{array}{c}
\mathcal{K}_{0^+}^{\#1} \quad \mathcal{K}_{0^+}^{\#2} \\
\mathcal{K}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \Upsilon_1 k^2 & \Upsilon_2 k^2 \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline \Upsilon_2 k^2 & \Upsilon_1 k^2 \\ \hline \end{array} \\
\mathcal{K}_{1^-}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline \frac{25 k^2 \binom{4}{\kappa_2}}{3} & -\frac{5}{3} \sqrt{5} k^2 \binom{4}{\kappa_2} \\ \hline \end{array} \\
\mathcal{K}_{1^-}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline -\frac{5}{3} \sqrt{5} k^2 \binom{4}{\kappa_2} & \frac{5 k^2 \binom{4}{\kappa_2}}{3} \\ \hline \end{array} \\
\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|} \hline 6 k^2 \binom{4}{\kappa_2} \\ \hline \end{array} \mathcal{K}_{3^+}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline 0 \\ \hline \end{array}
\end{array}$$

$$\begin{array}{c}
\mathcal{J}_{0^+}^{\#1} \quad \mathcal{J}_{0^+}^{\#2} \\
\mathcal{J}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \frac{\Upsilon_1}{24 k^2 \binom{4}{\kappa_2} \binom{4}{\kappa_1 + \kappa_2}} & -\frac{\Upsilon_2}{24 k^2 \binom{4}{\kappa_2} \binom{4}{\kappa_1 + \kappa_2}} \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline -\frac{\Upsilon_2}{24 k^2 \binom{4}{\kappa_2} \binom{4}{\kappa_1 + \kappa_2}} & \frac{\Upsilon_1}{24 k^2 \binom{4}{\kappa_2} \binom{4}{\kappa_1 + \kappa_2}} \\ \hline \end{array} \\
\mathcal{J}_{1^-}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline \frac{1}{12 k^2 \binom{4}{\kappa_2}} & -\frac{1}{12 \sqrt{5} k^2 \binom{4}{\kappa_2}} \\ \hline \end{array} \\
\mathcal{J}_{1^-}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline -\frac{1}{12 \sqrt{5} k^2 \binom{4}{\kappa_2}} & \frac{1}{60 k^2 \binom{4}{\kappa_2}} \\ \hline \end{array} \\
\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|} \hline \frac{1}{6 k^2 \binom{4}{\kappa_2}} \\ \hline \end{array} \mathcal{J}_{3^+}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline 0 \\ \hline \end{array}
\end{array}$$

Abbreviations used in matrices

$$\Upsilon_1 == \binom{4}{\kappa_1} + 7 \binom{4}{\kappa_2} \text{ \&\& } \Upsilon_2 == \binom{4}{\kappa_1} - 5 \binom{4}{\kappa_2}$$

Lagrangian

$$\binom{4}{\kappa_1} \partial_\beta \mathcal{K}_\chi{}^\delta \partial^\chi \mathcal{K}_\alpha{}^\beta + \binom{4}{\kappa_2} \partial_\chi \mathcal{K}_\beta{}^\delta \partial^\chi \mathcal{K}_\alpha{}^\beta + 18 \binom{4}{\kappa_2} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta - 12 \binom{4}{\kappa_2} \partial^\chi \mathcal{K}_\alpha{}^\beta \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta$$

Added source term(s):		$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{1^-}^{\#2\alpha} + \mathcal{J}_{1^-}^{\#1\alpha} == 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}^{\beta\chi}{}_\beta == \partial_\chi \partial^\chi \mathcal{J}^{\alpha\beta}{}_\beta$
$\mathcal{J}_{3^+}^{\#1\alpha\beta\chi} == 0$	7	$ \begin{aligned} & k^\delta k^\epsilon (3 \partial_\epsilon \partial^\chi \partial^\beta \partial^\alpha \mathcal{J}_\delta{}^\phi{}_\phi - \partial_\epsilon \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^{\chi\phi}{}_\phi - \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\beta\phi}{}_\phi - \partial_\epsilon \partial_\delta \partial_\epsilon \partial^\beta \mathcal{J}^{\alpha\phi}{}_\phi + 2 \partial_\phi \partial^\chi \partial^\beta \partial^\alpha \mathcal{J}_{\delta\epsilon}{}^\phi - \\ & 4 \partial_\phi \partial_\epsilon \partial^\beta \partial^\alpha \mathcal{J}^{\chi}{}_\delta{}^\phi - 4 \partial_\phi \partial_\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\beta}{}_\delta{}^\phi - 4 \partial_\phi \partial_\epsilon \partial^\chi \partial^\beta \mathcal{J}^\alpha{}_\delta{}^\phi + 5 \partial_\phi \partial_\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\phi} + 5 \partial_\phi \partial_\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\phi} + \\ & 5 \partial_\phi \partial_\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\phi} - 5 \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\alpha\beta\chi} - \eta^{\beta\chi} \partial_\nu \partial_\epsilon \partial_\delta \partial^\alpha \mathcal{J}_\phi{}^\nu - \eta^{\alpha\chi} \partial_\nu \partial_\epsilon \partial_\delta \partial^\beta \mathcal{J}_\phi{}^\nu - \eta^{\alpha\beta} \partial_\nu \partial_\epsilon \partial_\delta \partial^\chi \mathcal{J}_\phi{}^\nu + \\ & \eta^{\beta\chi} \partial_\nu \partial_\phi \partial_\epsilon \partial^\alpha \mathcal{J}_\delta{}^\phi{}_\nu + \eta^{\alpha\chi} \partial_\nu \partial_\phi \partial_\epsilon \partial^\beta \mathcal{J}_\delta{}^\phi{}_\nu + \eta^{\alpha\beta} \partial_\nu \partial_\phi \partial_\epsilon \partial^\chi \mathcal{J}_\delta{}^\phi{}_\nu - \eta^{\beta\chi} \partial_\nu \partial_\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\alpha\phi}{}_\nu - \eta^{\alpha\chi} \partial_\nu \partial_\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\beta\phi}{}_\nu - \\ & \eta^{\alpha\beta} \partial_\nu \partial_\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\chi\phi}{}_\nu + \eta^{\beta\chi} \partial_\nu \partial^\nu \partial_\epsilon \partial_\delta \mathcal{J}^{\alpha\phi}{}_\phi + \eta^{\alpha\chi} \partial_\nu \partial^\nu \partial_\epsilon \partial_\delta \mathcal{J}^{\beta\phi}{}_\phi + \eta^{\alpha\beta} \partial_\nu \partial^\nu \partial_\epsilon \partial_\delta \mathcal{J}^{\chi\phi}{}_\phi) == 0 \end{aligned} $
Total # constraint(s):	10	
Resolved unitarity condition(s):		True